

Process Cycle Efficiency

SLIBRARY 07 · SHEET 2 / 5

Value-add time / total time

PROFESSIONAL DEFINITION

Process Cycle Efficiency is the ratio of value-added time to total cycle time in a process, quantifying how much of a process actually creates value versus waiting and rework.

WHO DEVELOPED IT

A core Lean metric, formalised in Lean and Six Sigma practice building on the Toyota Production System.

WHY IT WAS DEVELOPED

It was developed to expose hidden waste — the large share of cycle time spent waiting — that conventional efficiency measures miss.

FIGURE 1 · PROCESS CYCLE EFFICIENCY

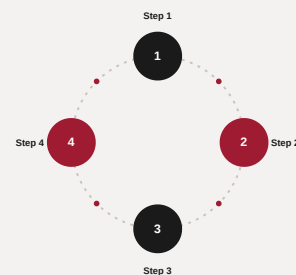


Figure 1. A schematic of the Process Cycle Efficiency as applied in BARBM312 (illustrative).

HOW IT IS USED

Value-added time is divided by total lead time; AI interventions are assessed by how much they raise the ratio rather than merely speed isolated steps.

WHEN IT IS USED

During process improvement and when targeting AI at genuine bottlenecks.

BY WHOM IT IS USED

Operations and continuous-improvement teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It focuses improvement on flow efficiency rather than local speed, ensuring AI is applied where it reduces total cycle time.

RELATED FRAMEWORKS IN THIS COURSE

Cost-to-Serve Curve · Slibrary 7

Lean Waste Lens (AI) · Slibrary 7

Throughput Accounting · Slibrary 7

Effort-Value Reallocation · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

George, M.L. (2002) Lean Six Sigma. New York: McGraw-Hill.

Womack, J.P. and Jones, D.T. (1996) Lean Thinking. New York: Simon & Schuster.